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    GitHub & Project Demo Link

Global Olympic Sports Performance Analysis

Project Report | Tableau-based sports analytics | Prepared by Keshav Trivedi | 27 October 2026

Executive Summary

This project organizes Olympic performance measures into an interactive Tableau workbook so a reader can compare country medal rankings, category averages, year-level medal trends, regional efficiency, regional medal distribution, host-nation labels and the relationship between investment score and medal output. The delivered workbook contains seven worksheets, two dashboards and four packaged Hyper extracts. The analysis is descriptive: the charts support exploration, but they do not prove why one country or region performs better.

Evidence and status note: the Tableau workbook and a five-minute walkthrough video are supplied. The workbook's structure has been inspected. Detailed row-level quality audit, true interactive response-time measurements, a public Tableau URL, a Tableau Story object, and executed Snowflake query logs were not present in the supplied artifacts and are identified as follow-up items rather than claimed as completed.

1. Introduction

1.1 Project Overview

The Global Olympic Sports Performance Analysis workbook brings several dimensions of Olympic results into a single analytical package. It uses the Athletes, Country Rankings, Games Summary and Medal Results extracts. The worksheets cover average medals by sport category, country medal totals/ranks, year-by-year total medals awarded, investment score versus medal output, average medal efficiency by region, medals across Olympic hosts and medal distribution by region.

1.2 Purpose

The purpose is to make comparative questions easier to explore, reduce manual inspection of separate tables, and support a concise discussion of visible patterns. It is intended as an exploratory dashboard, not a causal model or a definitive ranking of sporting quality independent of event counts and data scope.

2. Ideation Phase

2.1 Problem Statement

Students, researchers and sports administrators need a clear way to compare Olympic performance across countries, sports, regions and years. Raw tables and separate rankings make cross-view comparison slow and can encourage oversimplified conclusions when metric definitions or data coverage are unclear.

2.2 Empathy Map Canvas

- Says: “I need a quick, reliable way to compare Olympic performance.”
- Thinks: Raw tables are difficult to compare and measure definitions may differ.
- Does: Opens dashboards, filters categories or host groups, compares rankings and trends.
- Feels: Curious, but uncertain when source provenance or metric meaning is unclear.
- Pain points: Fragmented views, limited source notes, inconsistent labels and medal totals without context.
- Needs: Clear titles, interactive filters, transparent definitions and cautious interpretation.

2.3 Brainstorming and Prioritization

Candidate ideas included a country medal leaderboard, medal trends by Olympic year, category comparison, regional-efficiency analysis, investment-versus-medal scatter plot and host-nation comparison. The selected approach combines these high-feasibility, high-importance views across two dashboards.

3. Requirement Analysis

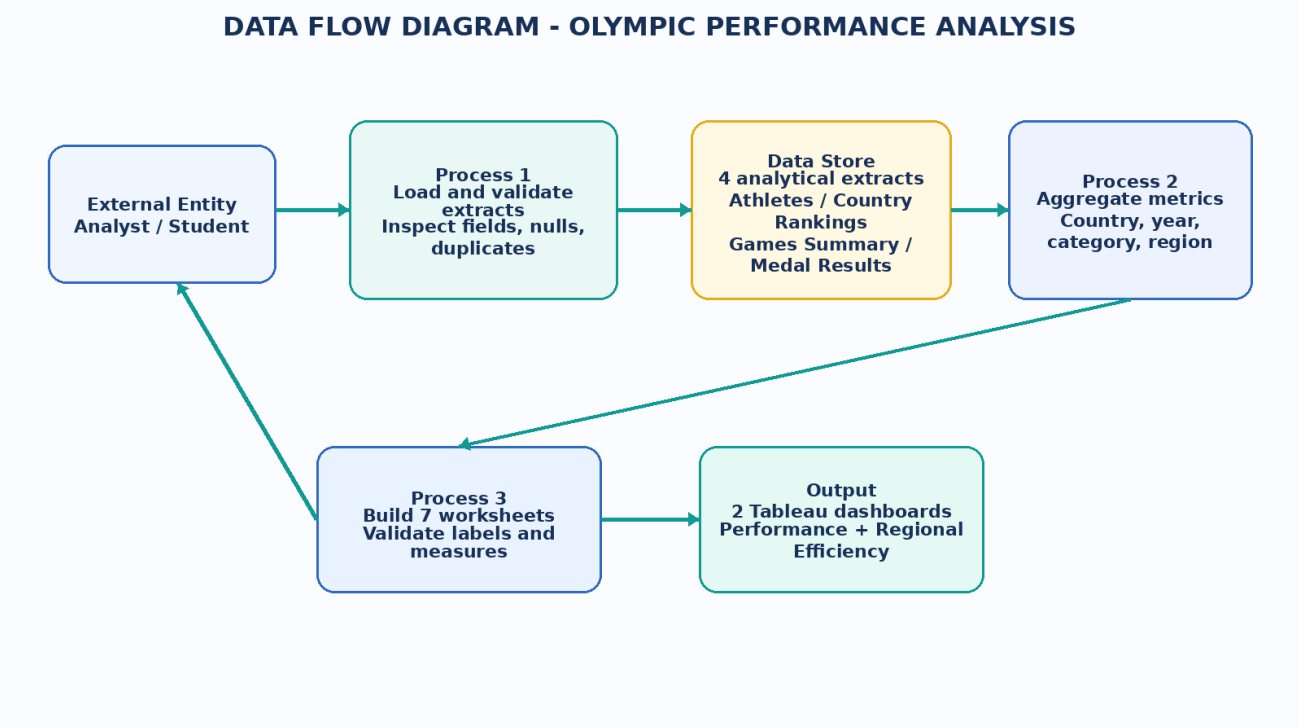
3.1 Customer Journey Map

Discover: open the packaged workbook and select a question. Explore: compare country/category/year/region views and use available filters. Act: summarize patterns in a report or demonstration. Improvement opportunities include source provenance, metric definitions, data-quality checks, richer tooltips and an embedded Tableau Story.

3.2 Solution Requirements

- Read the four packaged Hyper extracts and preserve field types and labels.
- Aggregate medal totals, year totals and category/region measures consistently.
- Provide seven visualizations grouped in two dashboards.
- Support country-rank range and host-advantage filtering where configured.
- Document metric meanings, data scope, source links and limitations.
- Allow a reproducible local workbook review; public publication is a separate step.

3.3 Data Flow Diagram



3.4 Technology Stack

Layer	Technology / current state
Data source	Four packaged Hyper extracts inside the .twbx workbook
Analysis and visuals	Tableau Desktop / Tableau Public compatible workbook
Optional QA	SQL or Python/pandas, if original source files are available
Sharing	Local TWBX now; publish to Tableau Public only after review
Snowflake	Optional extension only; no Snowflake load or query log was supplied

4. Project Design

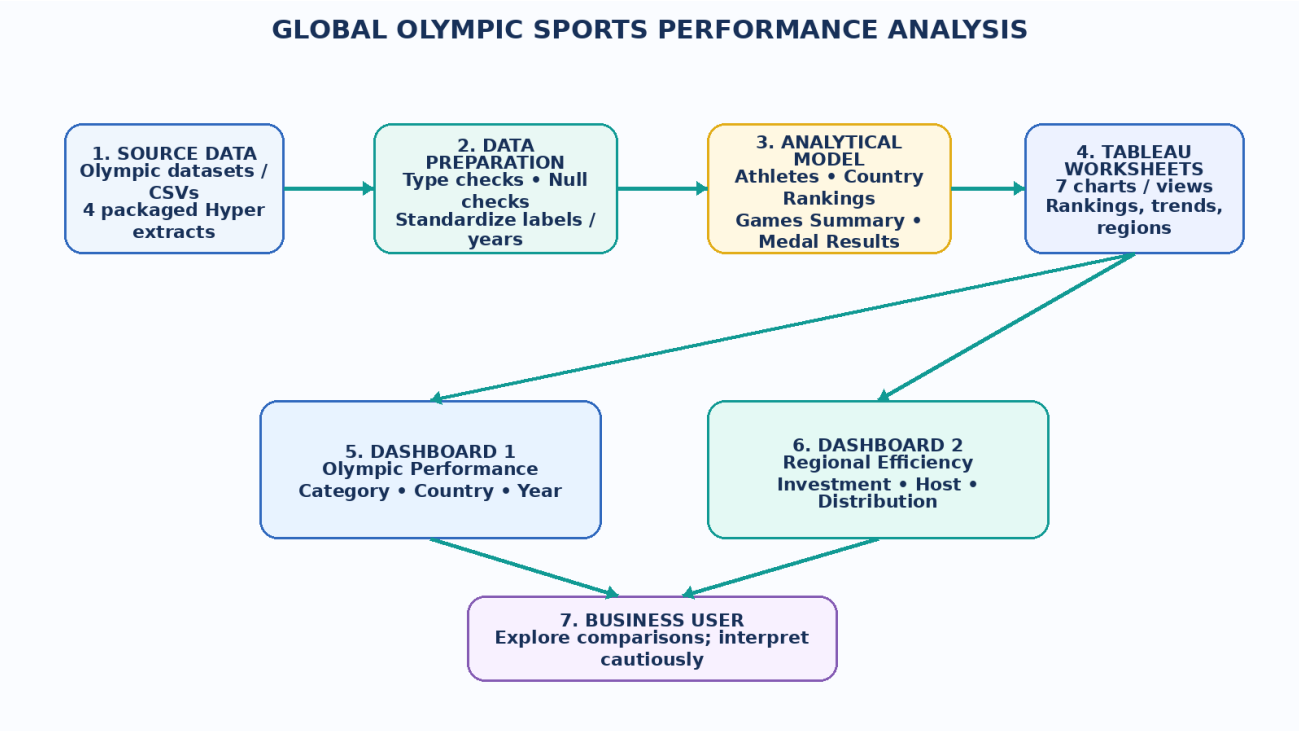
4.1 Problem-Solution Fit

The target users already rely on spreadsheets, static rankings and separate charts. A packaged workbook meets the need for a fast comparative view by presenting several measures in one place. Its value depends on trustworthy source data and clear measurement definitions; users remain constrained by limited provenance and the fact that medal counts can vary with event coverage.

4.2 Proposed Solution

The solution uses seven worksheets and two dashboard tabs. Dashboard 1 covers categories, country performance and year trend; Dashboard 2 covers investment-versus-medal output, regional efficiency, host-nation comparisons and regional medal distribution.

4.3 Solution Architecture

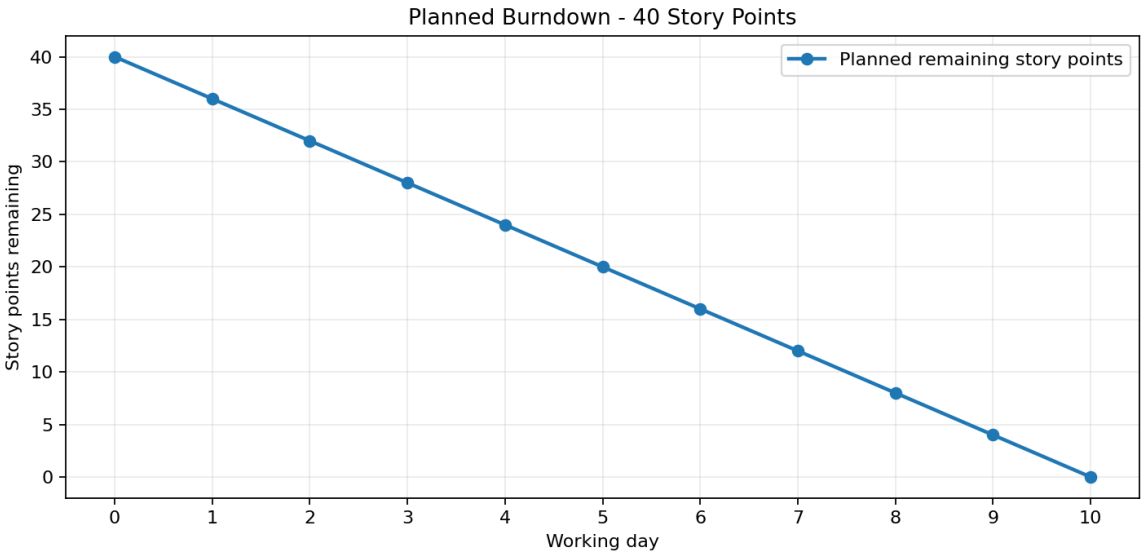


5. Project Planning and Scheduling

A four-sprint plan was drafted for final validation and documentation. Sprint 1 covers data/source review; Sprint 2 covers metric and regional review; Sprint 3 covers dashboard checks and the story outline; Sprint 4 covers performance evidence, documentation and submission packaging. Each sprint is planned at 20 story points over three working days (planning assumption, not measured past velocity).

Sprint	Planned focus	Points	Window
1	Data audit and core country/category/year views	20	28-30 Oct 2026
2	Regional, investment and host comparison review	20	31 Oct-2 Nov 2026
3	Dashboard review, storytelling and evidence	20	3-5 Nov 2026
4	Testing, documentation and packaging	20	6-8 Nov 2026

Illustrative planned burndown (not actual tracked progress):


Planned Burndown - 40 Story Points
Planned remaining story points
Story points remaining
Working day
0
5
10
15
20
25
30
35
40
0
1
2
3
4
5
6
7
8
9
10

6. Functional and Performance Testing

6.1 Tableau Workbook Checks

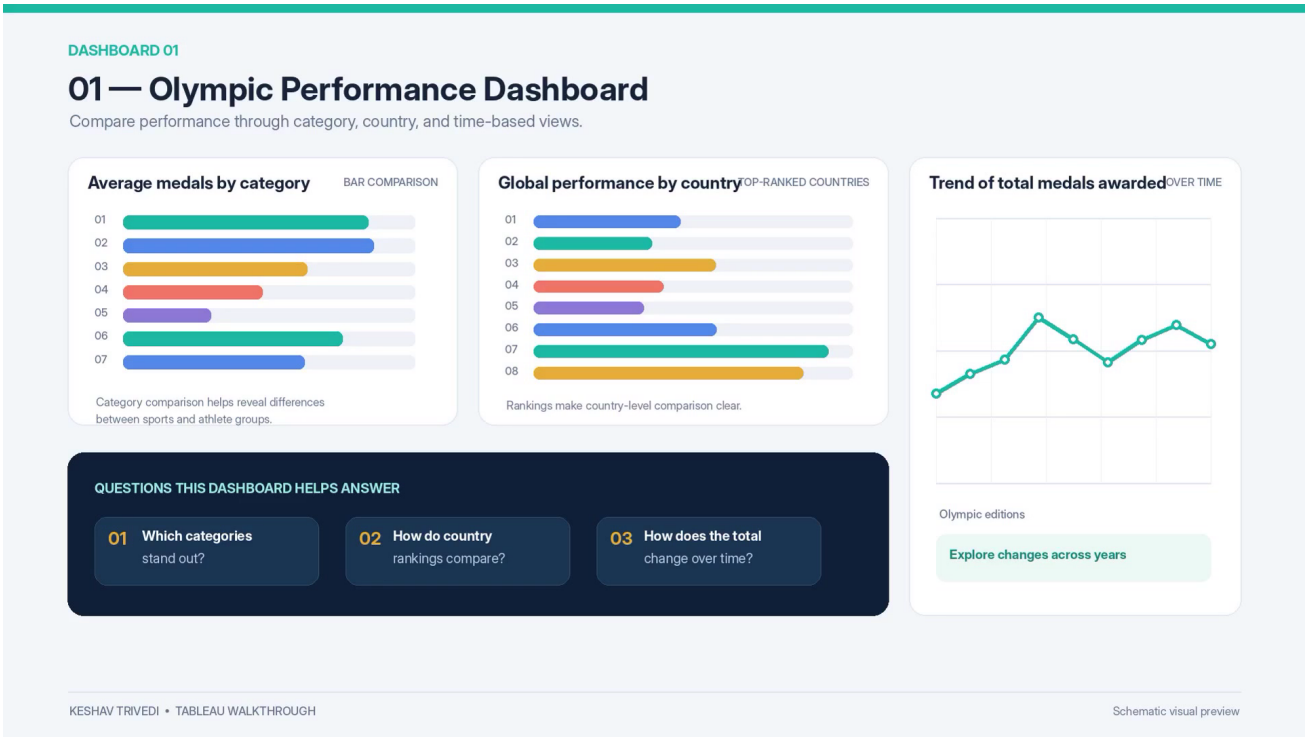
- Static inspection confirmed 4 packaged Hyper extract files, 7 worksheets and 2 dashboards.
- Two filter definitions are present in workbook XML: Overall Rank range and Host Advantage categorical filter.
- No custom calculation blocks were detected; views use aggregation of existing fields including Total Medals, Total Medals Awarded, Investment Score and Medal Efficiency Pct.
- Interactive open/render time, filter response time, row counts, and null/duplicate quality checks have not been measured in a live Tableau session.
- No Tableau Story element or public Tableau URL was found in the supplied workbook.

6.2 Snowflake Status

Snowflake-specific documents and a SQL reference script are included as optional implementation guidance. They are not executed results. The project package did not include a Snowflake database, warehouse, table-load history or query logs; no query execution times are claimed.

7. Results

Dashboard 1 - Olympic Performance

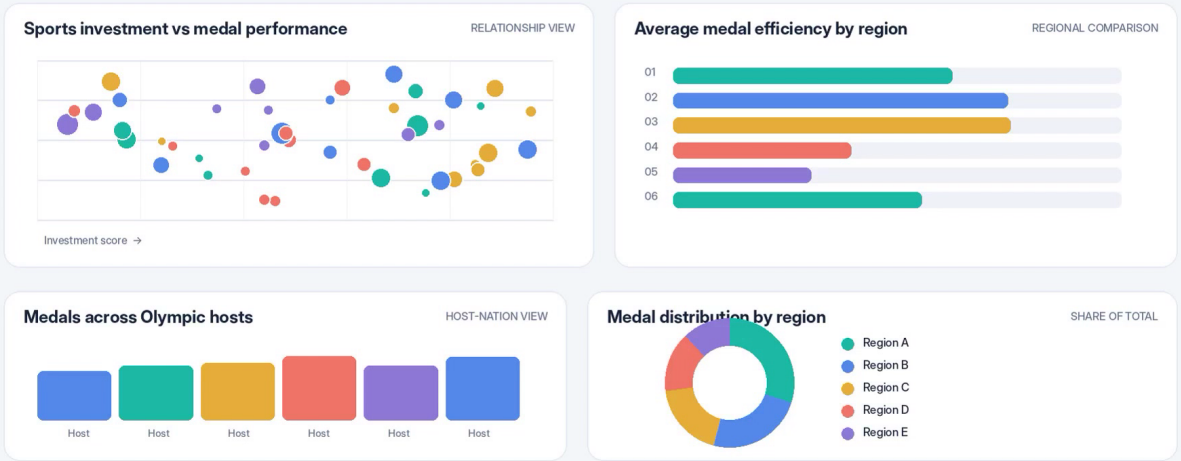


Dashboard 2 - Regional Efficiency

DASHBOARD 02

02 — Regional Efficiency Dashboard

Shift the focus from totals to relationships, regional comparisons, and host nations.



KESHAV TRIVEDI • TABLEAU WALKTHROUGH

Schematic visual preview

8. Advantages and Disadvantages

Advantages

- Combines country, category, time and regional comparisons in one workbook.
- Packaged extracts make the workbook easier to transfer and review locally.
- Two dashboards divide broad performance questions from regional/context comparisons.
- Visuals can support questions for further analysis and help users spot patterns.

Disadvantages / limitations

- Source-data URL, row-level data dictionary and full coverage description were not provided.
- Medal totals are not automatically normalized for number of events, athletes, or participation.
- Investment-versus-medal views do not establish causal effects.
- Public Tableau publication, actual Tableau Story points, and live performance/quality metrics remain outstanding.
- Snowflake execution is not evidenced and should not be presented as implemented.

9. Conclusion

The workbook offers a useful descriptive overview of Olympic performance through seven visualizations and two dashboards. Its strength is the ability to compare multiple dimensions and move between broad rankings, time trends, and regional context. Final numeric claims should be checked against the underlying extract values and metric definitions; the dashboard should be presented as exploratory rather than causal.

10. Future Scope

- Add a documented and licensed source dataset link with a data dictionary and clear coverage dates.
- Run row-level missing-value, duplicate and metric-consistency tests; record counts and outputs.
- Create and test Tableau Story points; publish a verified Tableau Public link if allowed.
- Add year/country/region filters where helpful and check dashboard filter scope.
- Optionally load validated data into Snowflake and record execution times before claiming warehouse-based testing.

11. Appendix

Files included: Tableau workbook (.twbx), five-minute demonstration video (.mp4), narration script (.txt), project phase templates filled in DOCX/PDF, and optional Snowflake SQL reference script. The original source data URL, GitHub URL and Tableau Public URL were not provided and are intentionally not fabricated.